Fairness Preservation under Computational Budgets in Probabilistic Decision Systems

ICT 5143: AFGML Capstone | Problem 9

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Research question and decision context

- Context and stakeholders: bank outreach scored by a Bayesian network under a compute limit; affected are customers, outreach teams, serving engineers and auditors.
- $Y = 1$: recorded term-deposit subscription; $D = 1$: selected for *hypothetical* pre-contact outreach (top 20% of scores).
- Question: at a fixed operation budget, does choosing the fallback by validation worst-group deviation from the exact decision give a smaller Δ_{EOpp} than a global fallback, utility-only selection and post-budget threshold adjustment?
- Hypothesis: yes, within 0.01 AP loss. Refuted if any Holm-adjusted paired 95% interval contains 0 or is positive.

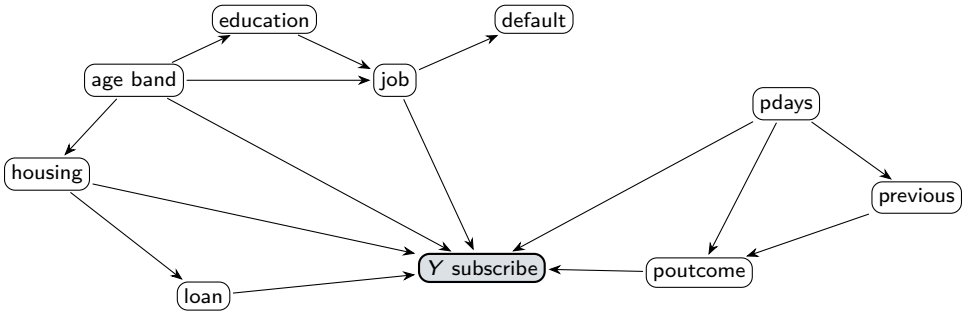
Closest work and proposed contribution

- Established result: reducing neural-network compute (fewer ReLUs) lowers accuracy more for minority groups (Das et al., ICML 2024); assumes neural models, accuracy parity.
- Open issue: no study found of budget-triggered *approximate PGM inference* and decision fairness (quick search only; novelty not claimed).
- Contribution: operation-count budget with evidence-dependent trips; simulation isolating group-dependent evidence availability, with a no-dependence control; fairness-aware fallback selection.
- Evidence needed: comparison with global fallback, utility-only selection and Hardt-style post-processing; ablations.

Data and evaluation population

- UCI Bank Marketing (dataset 222), `bank-additional-full.csv`: 41,188 contacts, 4,640 positives; $Y = 1$ iff `y = yes`; **no `duration`**.
- Protected: age band (< 30, 30–59, ≥ 60) $\times$ marital status (audit only, not a model input).
- Split (positional, date-ordered, no customer id): 24,712 / 8,238 / 8,238; positive rate 4.8% / 11.1% / 30.8%.
- Prediction-time inputs: age band, job, education, default, housing, loan, pdays bin, previous bin, poutcome; `unknown` $\Rightarrow$ unobserved evidence (any unknown: 13.6% of ages < 60, 20.9% of ≥ 60).
- Test groups: 7 supported (≥ 70 positives), 5 exploratory (1–8 positives, $n = 2$–34).

Graphical model and inference



$$p(\cdot) = p(A)p(E|A)p(J|A,E)p(F|J)p(H|A)p(L|H)p(P_d)p(P_r|P_d)p(O|P_r,P_d)\,p(Y|A,J,H,L,O,P_d)$$

- Query and algorithm: $P(Y{=}1 \mid e)$, e = observed variables; exact VE (min-fill, min-weight, min-degree) and clique tree (treewidth 6); cost = factor-entry mult + add.
- Assumptions: associational model, no latent variables; `unknown` treated as unobserved (A4).
- Validation: VE vs clique tree vs enumeration, max error 2.2×10^{-16} (tolerance 10^{-6}), 600 evidence patterns.

Fairness criterion and proposed method

- Primary fairness criterion: $\Delta_{\mathrm{EOpp}} = \max_g t_g - \min_g t_g$, $t_g = P(D{=}1 \mid Y{=}1, G{=}g)$; a missed responder is the harm to the group.
- Primary utility and allowed loss: average precision (AP), loss ≤ 0.01 vs exact (fixed before test).
- Proposed change: if exact elimination would exceed budget B operations, answer by table lookup; choose $f^\star = \arg\min_f \max_g \mathrm{flip}_g(f)$ s.t. validation AP loss ≤ 0.01.
- Protected attribute at prediction time: age band is a model variable (yes); routing is group-unaware; marital status unused.
- Diagnostics: Δ_{DP}, Δ_{EOdds}, fallback-rate gap, decision flips, log loss, Brier.

Experimental protocol

- Baselines: exact; global fallback; threshold adjustment after budget; fair-adjusted exact; logistic regression; gradient-boosted trees.
- Matched information and tuning: same variables and splits; thresholds, tables, selection from train/validation only; budget = natural validation op-count quantile (primary 75%: 11 ops).
- Selection: capacity 0.2; test protocol frozen in `docs/protocol.md`.
- Runs and intervals: 5 seeds (masks, networks, GBT); paired moving-block bootstrap, 1,000 replicates, blocks of 50 date-ordered rows; Holm over 3 comparisons.
- Mechanism tests: exact agreement; synthetic networks (width $\times$ availability, with control); ablations A1–A4; robustness R1–R6.

Main result and uncertainty

Method	AP ↑	Δ_{EOpp} ↓	Worst-group flip ↓	Log loss / Brier ↓
Exact (reference)	0.421	0.465	0	0.709 / 0.223
Global fallback (baseline)	0.350	0.303	0.303	0.852 / 0.260
Proposed	0.421	0.423	0.038	0.710 / 0.223
Utility-only (A1)	0.354	0.717	0.198	0.825 / 0.254
Threshold adj.	0.350	0.317	0.319	0.852 / 0.260

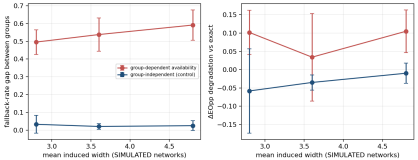
Natural workload, $n = 8{,}238$, 11 ops, capacity 0.2; Δ_{EOpp} over 7 supported groups.

Paired Δ_{EOpp} effect, proposed − comparator (negative favours proposed), 95% interval: global $+0.120$ $[0.015, 0.226]$, Holm $p = 0.036$; utility-only -0.294 $[-0.370, -0.210]$, $p = 0.006$; threshold adj. $+0.106$ $[-0.006, 0.251]$, $p = 0.080$.

Answer: hypothesis refuted. Proposed keeps exact decisions (98.3% agreement, AP loss 0), but global and threshold baselines reach lower Δ_{EOpp} only by levelling down (AP -0.071).

Ablation, robustness and failure case

- Ablation A1 (no fairness-aware selection): worst flip 0.038 $\rightarrow$ 0.198, Δ_{EOpp} 0.294 worse. A3 (no simplification): fallbacks 20.1% $\rightarrow$ 26.2%, AP 0.421 $\rightarrow$ 0.356. A4 (258 ops): no fallbacks, exact behaviour.
- Robustness: capacity 0.1/0.3 and elimination orders keep the pattern; dense / sparse networks raise proposed worst flip to 0.206 / 0.133.
- Failure case (synthetic dependent masks, exploratory): proposed picks the age-based fallback in 3 of 5 seeds, AP loss 0.046, Δ_{EOpp} $+0.493$ $[0.397, 0.569]$ vs exact; group-unaware variant loses 0.003 AP.
- Interpretation: bootstrap intervals are sampling uncertainty of fixed models; seed and scenario variation is separate; the simulation is not customer fact.



SIMULATED networks: fallback-rate gap 0.542 (group-dependent) vs 0.026 (control).

Conclusion and reproducibility

- Conclusion: fairness-aware selection preserves exact decisions at a tight budget, but does not give the smallest Δ_{EOpp}; the confirmatory hypothesis is refuted.
- Limits and ethics: base rate drifts 4.8% $\rightarrow$ 30.8% (test selection 38.3% vs 20%); no customer ids; contacted-only sample; 5 exploratory groups; simulations labelled.
- Reproduce: `scripts/run_experiments.py -mode real`, `evaluate.py -out results`, `make_results.py -out results -mode real`; `configs/main.yaml`; commit `e2350bcd6d58`.
- Shreeya Kamath: implementation (inference, baselines, experiments, notebook) and slide alignment.
- Misha: debugging, report writing and slides. Each cross-checked the other's work (`docs/contributions.md`).

Backup: mandatory evidence and references

ID	Requirement	Artifact (`results/metrics/`)	Status
C-GM	VE = clique tree = enumeration	`exp1_exact_agreement.json`, `cgm.json`	complete
C-STAT	Paired block bootstrap, Holm	`bootstrap.csv`	complete
Exp 2	Cost, peak factor, memory	`exp2_complexity.csv`	complete
Exp 3–4	Budgets, matched utility, baselines	`main_results.csv`, `group_table.csv`, `predictive_baselines.csv`	complete
Exp 5	Synthetic networks with control	`exp5_aggregate.csv`, `exp5b_factorial.csv`	complete
A1–A4, R1–R6	Ablations, robustness	`main_results.csv` (tags)	complete
Latency	Wall-clock, separate	`latency.csv`	complete

Das, Romanelli, Fioretto, ICML 2024 (arXiv:2402.03629); Hardt, Price, Srebro, NeurIPS 2016 (arXiv:1610.02413); Moro, Rita, Cortez, UCI dataset 222; Koller & Friedman, *Probabilistic Graphical Models*, MIT Press 2009. Full table: `docs/related_work.md`; manifest: `results/manifest.csv`.

Backup: inference and statistical validation

- Factor scopes: every CPD normalises over its child (checked). Maximal cliques (min-fill): $\{A, E, J\}$ 288 entries; $\{A, H, J, L, P_d, Y\}$ 3,888; $\{F, J\}$ 36; $\{P_d, O, P_r\}$ 12; treewidth 6.
- Exact agreement: max $|\mathrm{VE} - \mathrm{CT}| = 2.2 \times 10^{-16}$, vs enumeration 1.1×10^{-16} (tolerance 10^{-6}); 20 random 8-node networks $\times$ 10 queries $\times$ 3 heuristics also pass.
- Cost (natural evidence): VE mean 24.7 ops, peak factor 96; clique-tree calibration 44,790 ops per evidence (+29,337 preprocessing); wall-clock 0.225 vs 1.09 ms per query.
- Test denominators (positives / n): 30–59 married 932/3,152; <30 single 502/1,525; 30–59 single 451/1,631; ≥ 60 married 309/659; 30–59 divorced 176/650; ≥ 60 divorced 70/160; <30 married 79/384. Exploratory: <30 divorced 6/34; ≥ 60 single 8/25; 30–59 unknown 5/14; two unknown cells 1/2.
- Bootstrap: moving blocks of 50, 1,000 replicates, 95% percentile, statistic Δ_{EOpp} over the 7 supported groups; Holm family = 3 comparisons (natural workload). All-group sensitivity intervals are wide (e.g. vs global -0.167 $[-0.385, 0.554]$).

Backup: demonstration and individual defence

- Command: `python scripts/evaluate.py -out results && python scripts/make_results.py -out results -mode real` regenerates the bootstrap table and main table from saved runs (about 40 s); the full run took 1,307 s (Python 3.13, Linux).
- Saved trace (most expensive natural query; observed: age 30–59, default no, housing no, loan no, poutcome failure): min-fill eliminates previous_bin (6 ops), education (180), job (140), pdays_bin (8); total 339 ops; $P(Y{=}1 \mid e) = 0.1683$.
- Same posterior by min-degree order (339 ops) and by clique-tree calibration (41,325 ops); order changes cost, never the posterior.
- Be ready to: trace one operation count; justify Δ_{EOpp} and why levelling down matters; critique the claim given base-rate drift and tiny groups.